M1. Library Membership Foundation & Batch Enrollment

Category B Practice - Problem 1 (Basic)

Approach

LibraryMember performs all member ID validation; enrollBatch relies on constructor exceptions to count rejected entries. StudentMember extends LibraryMember directly and uses super(...) for common data.

Java Solution

```
class LibraryMember {
    protected final String memberId; protected final int borrowLimit;
    private int booksBorrowed;
    public LibraryMember(String memberId, int borrowLimit) {
        if (memberId == null || memberId.trim().length() < 4)
            throw new IllegalArgumentException("Invalid member ID");
        if (borrowLimit <= 0) throw new IllegalArgumentException("Limit must be positive");
        this.memberId=memberId; this.borrowLimit=borrowLimit;
    }
    public void borrowBook() {
        if (booksBorrowed >= borrowLimit) throw new IllegalStateException("Borrow limit reached");
        booksBorrowed++;
    }
    public int getBooksBorrowed() { return booksBorrowed; }
}
class StudentMember extends LibraryMember {
    private final String course;
    public StudentMember(String id,int limit,String course) { super(id,limit); this.course=course; }
}
class Enrollment {
    static String enrollBatch(String[] ids,int limit) {
        int enrolled=0,rejected=0;
        for(String id:ids) {
            try { new LibraryMember(id,limit); enrolled++; }
            catch(IllegalArgumentException ex) { rejected++; }
        }
        return "Enrolled: " + enrolled + " | Rejected: " + rejected;
    }
    public static void main(String[] args) {
        System.out.println(enrollBatch(new String[]{"STU1","LB1","STU2"," ","STU3"},3));
    }
}
```

Sample Output

```
Enrolled: 3 | Rejected: 2
```